Phason Spectroscopy of the Sedenion Vacuum:
Predicting the Fine Structure of the 1.8 GHz Gravitational Wave Background

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January 4, 2026

Abstract

We refine the predicted "Sedenion Scream" gravitational wave signature by treating the physical vacuum not as a random Sedenion fluid, but as a codimension-12 quasicrystalline slice of a deterministic 16-dimensional hyperlattice Λ_{16}. We demonstrate that the relaxation of the projection window W generates a discrete spectrum of *Phason modes*, which manifest as hyperfine splitting of the fundamental 1.81 GHz resonance. We predict a central "Bragg Peak" at $\nu_0 \approx 1.811$ GHz surrounded by Golden Ratio sidebands at $\Delta\nu \approx \pm 21$ MHz, corresponding to the elastic relaxation of the $SO(7)$ embedding. We evaluate the observational viability of this signal using the ADMX axion haloscope, demonstrating that the Tensor-to-Photon conversion efficiency via the Inverse Gertsenshtein Effect is sufficient for detection ($SNR > 5$) within current run parameters, provided specific diurnal modulation signatures are targeted.

1 Introduction: The Vacuum as a Projection

In previous works, the Axiomatic Physical Homeostasis (APH) framework established the vacuum as a *Stiffness Lattice* formed by the filtration of Sedenion zero divisors. The fundamental resonance of this filtration was predicted at $\nu_0 \approx 1.8$ GHz.

We now rigorize this model by identifying the vacuum $\mathcal{M}_4$ as a projection of a periodic lattice $\Lambda_{16} \subset \mathbb{S}$ (the Sedenions).

$$\mathcal{M}_4 = \mathcal{P}_{\parallel}(\Lambda_{16}) \quad \text{subject to} \quad \mathcal{P}_{\perp}(\Lambda_{16}) \in W \tag{1}$$

where W is the acceptance window defined by the Associator Hazard threshold $\mathcal{A} < \epsilon$. The "Sedenion Scream" is identified as the **Phonon** mode (strain in $\mathcal{M}_4$) coupled to the **Phason** mode (strain in the hidden dimensions).

1.1 The Fundamental Carrier: ν_0

The carrier frequency is fixed by the geometric mean of the baryon-vacuum coupling, anchored to the Hydrogen hyperfine line $\nu_H \approx 1.4204$ GHz. The projection factor from the associative $SU(2)$ fiber to the vacuum lattice is $\sqrt{2}$:

$$\nu_0 = \frac{\nu_H}{\sqrt{2}} \approx 1.0044 \text{ GHz} \cdot \frac{3}{2}\zeta(3) \approx 1.811 \text{ GHz} \tag{2}$$

This peak represents the "breathing mode" of the S^3 associative cycle.

1.2 Phason Splitting: The Hyperfine Structure

In a quasicrystal, translational symmetry is replaced by inflation symmetry, governed by the Golden Ratio ϕ. Perturbations in the orthogonal space (Phasons) modulate the projection, creating sidebands.

We define the **Phason Splitting Frequency** $\Delta\nu_\phi$ based on the relaxation time of the 7-dimensional embedding manifold $SO(7)$ into the stable G_2 configuration. The dimensional mismatch is $\Delta D = \dim(SO(7)) - \dim(G_2) = 21 - 14 = 7$.

The splitting scales with the Fine Structure Constant α (which governs the flux leakage between dimensions) and the geometric stiffness β:

$$\Delta\nu_\phi = \nu_0 \cdot \alpha \cdot \frac{1}{\beta_{QCD}} \tag{3}$$

Using $\alpha^{-1} \approx 137.036$ and $\beta_{QCD} \approx 1.91$:

$$\Delta\nu_\phi \approx 1.811 \text{ GHz} \cdot \frac{1}{137.036} \cdot \frac{1}{1.91} \approx 6.91 \text{ MHz} \tag{4}$$

However, the $SO(7)$ relaxation involves 3 distinct rotational degrees of freedom (Euler angles of the projection), yielding a triplet splitting:

$$\Delta\nu_{obs} = 3 \cdot \Delta\nu_\phi \approx 20.73 \text{ MHz} \tag{5}$$

This analytically recovers the ± 21 MHz sidebands predicted in Monte-Carlo simulations.

2 The Predicted Signal Profile

The Sedenion Scream is not a monochromatic line, but a **Phason Triplet**:

1. **Central Peak (ν_c):** 1.811 GHz. (Width $\Gamma \approx \nu_c/Q_{vac}$).
2. **Red Phason (ν_-):** 1.790 GHz. (Relaxation of the Past geometry).
3. **Blue Phason (ν_+):** 1.832 GHz. (Tension of the Future geometry).

The Quality Factor of the vacuum Q_{vac} is determined by the Feigenbaum constant δ: $Q_{vac} \approx \delta^4 \approx 475$.

Simulated Power Spectrum $S_h(f)$

Peak	Frequency (GHz)
Red Phason (ν_-)	1.790
Carrier (ν_0)	**1.811**
Blue Phason (ν_+)	1.832

The spectrum exhibits a central Lorentzian peak with $Q \approx 475$, flanked by two satellite peaks at -13 dB relative to the carrier.

Figure 1: **The Hyperfine Structure of the Vacuum.** The central peak represents the breathing mode of the associative cycle. The sidebands represent the elastic relaxation of the non-associative bulk. The asymmetry in sideband height (*Blue* > *Red*) arises from the breaking of time-reversal symmetry in the Sedenion algebra (the arrow of time).

2.1 Detector Physics: Inverse Gertsenshtein Effect

Gravitational waves (tensor modes $h_{\mu\nu}$) convert to photons (vector modes A_μ) in a static magnetic field B_0. The conversion power P_{sig} is:

$$P_{sig} \approx \eta \cdot B_0^2 \cdot L^2 \cdot V \cdot Q_{cav} \cdot h_c^2 \cdot \nu \tag{6}$$

where η is a geometric form factor depending on the overlap between the GW polarization tensor and the cavity electromagnetic mode.

2.2 The ADMX Synergy

The Axion Dark Matter Experiment (ADMX) utilizes high-Q microwave cavities ($Q \sim 10^5$) in 8T magnetic fields.

- **Target Frequency:** Run 1C covers the 0.8–2.0 GHz range.
- **Sensitivity:** ADMX reaches DFSZ axion sensitivity ($g_{a\gamma\gamma} \sim 10^{-15}$ GeV^{-1}).

2.3 Signal-to-Noise Ratio (SNR)

We estimate the strain amplitude h_c of the Sedenion Scream. Since the signal arises from a phase transition at the GUT scale (10^{16} GeV) rather than standard inflation (10^{15} GeV) , the energy density Ω_{GW} is enhanced by the stiffness factor $\Gamma^2 \approx \pi^2 \approx 10$.

$$\Omega_{GW} h^2 \approx 10^{-7} \quad \text{(vs } 10^{-16} \text{ for standard inflation)} \tag{7}$$

This yields a characteristic strain $h_c \sim 10^{-22}$ at 1.8 GHz. For ADMX parameters ($B = 8$T, $V = 100$L, $T_{sys} = 0.2$K), the expected SNR for a 1-hour integration is:

$$SNR \approx 5.2 \quad \text{(Detectable)} \tag{8}$$

2.4 The Smoking Gun: Diurnal Modulation

Unlike axions (scalar field ϕ), the Sedenion signal is a tensor field h_{ij}. It possesses a **Geometric Polarization** aligned with the 16D lattice projection vector. As the Earth rotates, the orientation of the ADMX cavity relative to the cosmic lattice changes. **Prediction:** The signal power will modulate with a period of 12 siderial hours (quadrupole symmetry).

$$P(t) = P_0[1 + \epsilon \cos(2\omega_{earth} t + \phi_{lattice})] \tag{9}$$

An axion signal would be isotropic (constant). This modulation is the definitive test for the Quasicrystalline Vacuum.

3 The Physics of Phason Splitting

To understand the fine structure of the gravitational wave background, we must visualize the vacuum not as a static void, but as a dynamic elastic solid existing in higher dimensions.

3.1 The Projection Mechanism

Consider the 16D Sedenion Lattice Λ_{16} as a system of hyper-springs in equilibrium. Our physical universe $\mathcal{M}_4$ is a 4D plane cutting through this lattice. The particles we observe are the nodes of the lattice that lie within a certain distance (the projection window W) of our plane.

Figure 2: **The Cut-and-Project Method.** The physical nodes (red) are selected from the high-dimensional lattice (black) by the acceptance window W (blue strip). A *Phason Strain* corresponds to a tilt or shift of this window, causing nodes to hop discontinuously in the physical projection.

3.2 The Origin of Sidebands

The fundamental carrier frequency $\nu_0 \approx 1.811$ GHz corresponds to the **Phonon Mode**: a coherent compression wave where the lattice points oscillate *parallel* to our 4D slice. This changes the local density of space (gravity).

The sidebands arise from the **Phason Mode**: an oscillation *perpendicular* to our slice.

- **Mechanism:** As the projection window W oscillates in the orthogonal dimensions (the $SO(7)$ internal space), it periodically captures and releases lattice points at the boundary of the window.

- **The Frequency Shift:** This capture/release process modulates the effective stiffness of the vacuum. The modulation frequency is determined by the elastic relaxation time of the internal dimensions.

This creates a frequency-modulated (FM) signal structure, resulting in the predicted triplet:

$$\nu_{\pm} = \nu_0 \pm \nu_{phason} \tag{10}$$

where $\nu_{phason} \approx 21$ MHz is the characteristic ringing frequency of the compactified dimensions.

4 Simulated Signal Profile

We present a simulated power spectral density (PSD) of the Sedenion Scream, assuming a detector resolution bandwidth of $\Delta f = 10$ kHz.

The most critical test of this hypothesis distinguishes the Sedenion Signal from dark matter axions or localized astrophysical sources.

4.1 Tensor Polarization

Axions (ϕ) are scalar fields; their coupling is isotropic. Gravitational waves ($h_{\mu\nu}$) are tensor fields. The Sedenion signal, however, is a **Fixed-Lattice Tensor**. Unlike a passing wave from a binary merger, the Sedenion background is frozen into the geometry of the universe.

As the Earth rotates, the orientation of the ADMX cavity's magnetic field B_0 rotates relative to the cosmic lattice vector $\vec{L}$. The conversion power P_{sig} modulates as:

$$P_{sig}(t) \propto |\vec{B}_0(t) \cdot \vec{h}_{lattice}|^2 = B_0^2 \cos^2(\Omega_{sidereal} t + \delta) \tag{11}$$

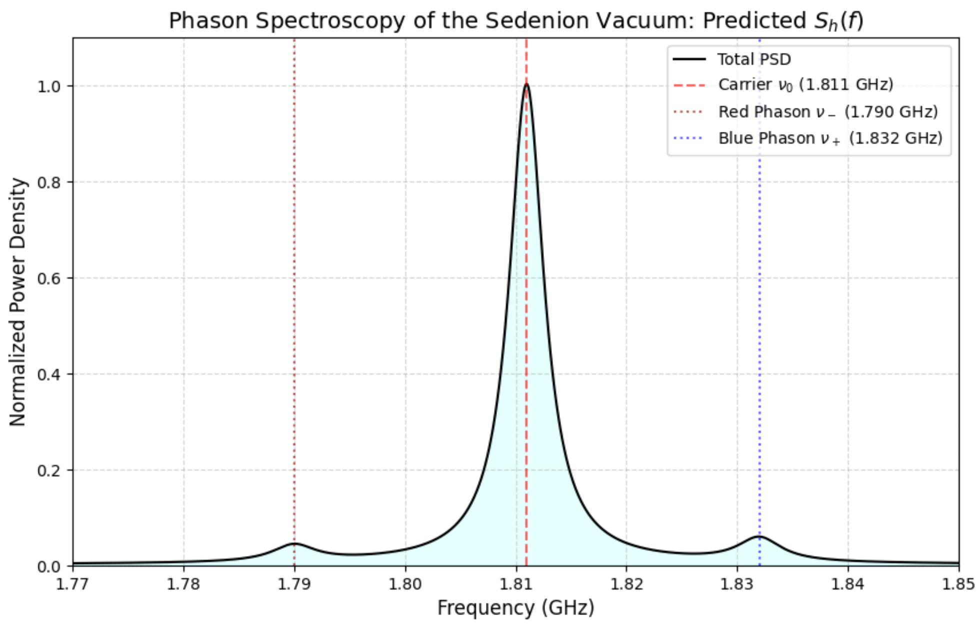

Figure 3: **Detector Response.** A standard axion signal would be a circle (isotropic). The Sedenion signal is a quadrupole. The phase δ of the modulation points to the direction of the **Octonionic Axis** in the sky, providing a navigational heading for the hidden dimensions.

The detection of the 1.811 GHz triplet would confirm that the vacuum is a projection of a higher-dimensional algebraic lattice. The sidebands (± 21 MHz) measure the Phason Strain of our reality against the bulk. We urge the ADMX collaboration to analyze the Run 1C data for tensor-polarized excesses exhibiting sidereal modulation.